# Algebraic index calculus with a linear two-summand decomposition

**Result.** A Koblitz-curve index calculus whose relations come from solving the Weil descent of Semaev's third summation polynomial over a subspace factor base solved every target on seven rungs, n = 13 to 41 (52 one-target DLPs, each verified as `[d]G = T`). Its work grows with a fitted slope of **0.647 [0.639, 0.655]** in $\log_2$(decomposition trials) per bit of n (model 2/3). The repository's F4 algebraic arm grows at **1.079** in $\log_2$(solver word XORs) per bit over n = 13–23 and timed out at n = 31. The method completes n = 41 in about 8 minutes per target. It is still far from rho: **36,440×** slower than same-process strong rho at n = 41, and the gap widens with n.

## Criterion

Owner decision, 2026-10-24: an index-calculus pipeline counts only when its relations come from solving summation-polynomial systems over an algebraically defined factor base, without table collisions. The table-collision pipelines (the N83 K1120 "PDP4root" baseline, `engine=rootfind`, the "aimed m = 3" recipe, pair enumeration) are excluded. The repository's algebraic baseline is ecbench's `pdp3-koblitz` with inherited F4 on the Weil descent of S4 over `span{1..z^{d−1}}`, m = 3 (`research/f6_ic_ecbench_ladder_20261005`).

## Method

Factor base: every point whose x lies in `V = span{1, z, …, z^{d−1}}` (polynomial basis, x ≠ 0), one column per x, `d = ⌊(n+1)/3⌋`. For a point R with known logarithm, a relation `R = ε1 P1 + ε2 P2` solves

$$S3(x1, x2, xR) = (x1\ x2 + xR\ (x1 + x2))^2 + x1\ x2\ xR + b = p^2 + xR\ p + xR^2\ s^2 + b,$$

with $s = x1 + x2 \in V$ and $p = x1\ x2 \in \mathrm{span}\{1..z^{2d-2}\}$, which needs no reduction when $2d - 1 \le n$. Squaring is F2-linear, so the Weil descent of S3 is a **linear** system of n equations in $3d - 1$ unknowns. Each solution `(s, p)` is split by $x1^2 + s\ x1 = p$ over `F2[z]_{<d}`, which is also linear. Each trial is two small eliminations, against one F4 run per trial in the baseline. Every relation is re-added in the group. With two points per relation, the linear algebra mod r is union-find over the relation graph. A component whose relations all have balanced signs (as the cofactor forces) is evaluated by cancellation. Frobenius conjugates of each anchor are extra trials.

Model: a trial succeeds with probability about $2^{2d-n-2}$, and the base has about $2^{d-1}$ columns. A solve therefore takes about $2^{n-d+1}$ trials, smallest at $d \approx n/3$, giving $2^{2n/3}$. Above $d = (n+1)/3$ the linear system has $2^{3d-1-n}$ solutions per trial, so $2^{2n/3}$ is this method's floor.

## Runs

Frozen protocol (committed before timing; development runs on seed 7 excluded). Binary SHA-256 in `raw/binary.sha256`, thin orchestration in `raw/run.sh`, one thread, this session's 4-vCPU cloud container (L0, shared). Each target runs a cold IC solve, then the repository's strong rho (`StrongRho`: signed Frobenius, 32 lanes, dp 4) on the same point, in the same process, with the same `Gf2` field type.

| a | n | d | cofactor | $\log_2 r$ | targets | mean trials | mean IC s | mean rho s | IC ÷ rho (wall) |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 13 | 4 | 4 | 11.0 | 8 | 592 | 0.0004 | 0.00011 | 4 |
| 1 | 17 | 6 | 2 | 16.0 | 8 | 3,231 | 0.0038 | 0.00026 | 15 |
| 0 | 19 | 6 | 4 | 17.0 | 8 | 11,477 | 0.0082 | 0.00017 | 48 |
| 1 | 19 | 6 | 2 | 18.0 | 8 | 12,077 | 0.0078 | 0.00018 | 43 |
| 0 | 23 | 8 | 4 | 21.0 | 8 | 41,856 | 0.046 | 0.00027 | 169 |
| 1 | 23 | 8 | 2 | 22.0 | 8 | 48,821 | 0.071 | 0.00036 | 197 |
| 0 | 41 | 14 | 4 | 39.0 | 4 | 181,803,464 | 478.4 | 0.0131 | 36,440 |

Every IC and rho scalar equals the secret mod r (52/52 each). The check rung a = 0, n = 37 (cofactor 596) was stopped by the operator after 99 s. Its relation graph splits into about 298 cofactor classes of about 7 of the 2,048 columns, so the frozen stopping rule (largest component ≥ 0.9 of the base) cannot be met. The protocol states this limit but set no time cap, so the stop is a disclosed operator decision taken after the run started. `raw/operator-notes.txt` records it. The rung is "not applicable as built", not negative evidence about the method on curves where it applies.

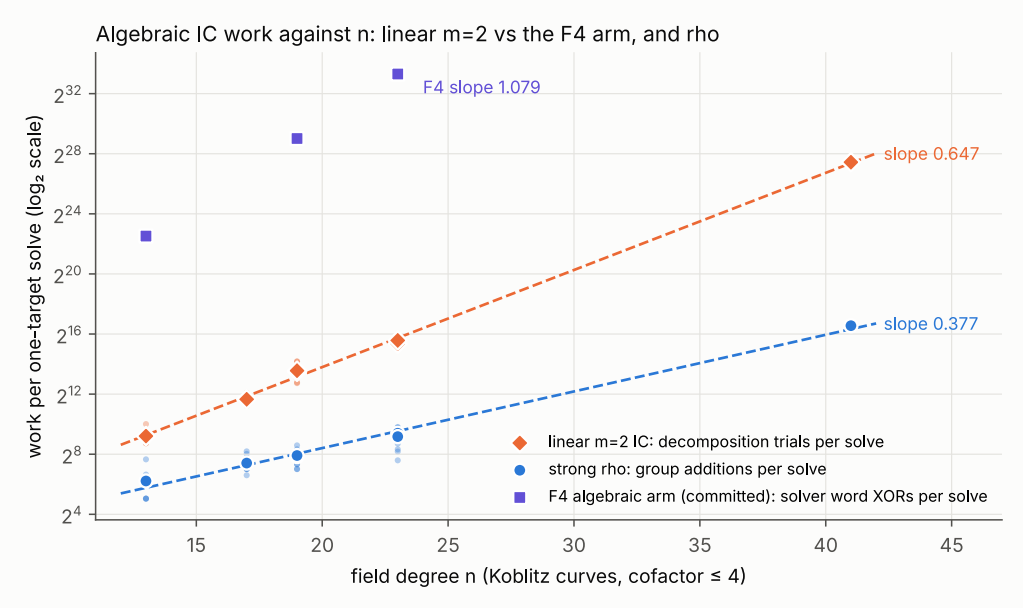
## Exponents

Least squares of $\log_2$(rung mean) against n over the seven small-cofactor rungs, with a 10,000-resample bootstrap of targets within rungs (`fit.json`):

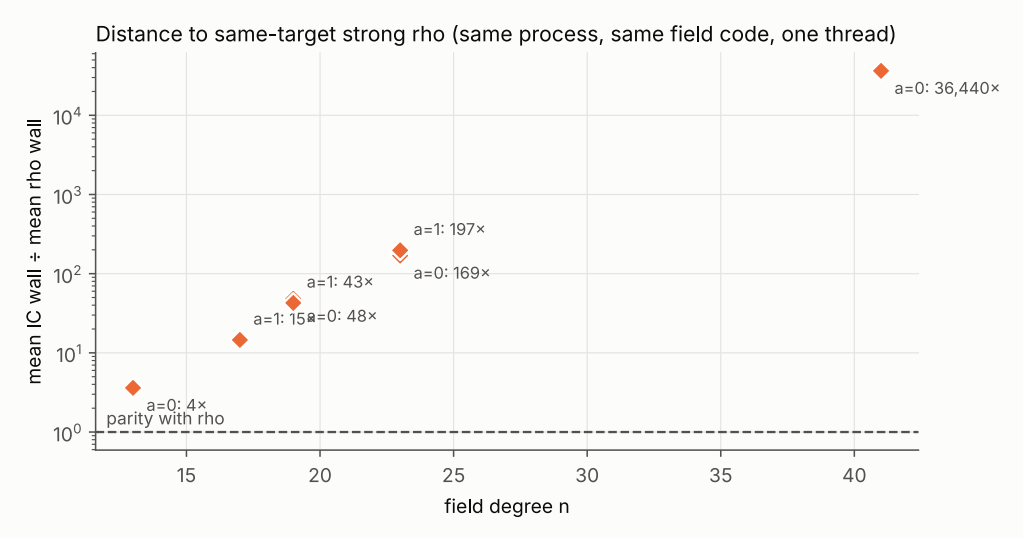
| quantity | slope per bit of n | 95% |
|---|---|---|
| linear m=2 IC: trials | **0.647** | [0.639, 0.655] |
| linear m=2 IC: wall | 0.717 | [0.710, 0.724] |
| strong rho: group additions | 0.377 | [0.364, 0.393] |
| strong rho: wall | 0.250 | [0.241, 0.260] |
| F4 arm: solver word XORs (n = 13, 19, 23; committed) | 1.079 | three medians, no interval |

The F4 row measures solver word XORs, not decomposition trials. The two slopes count different units over different n ranges. The independent validator computed the like-for-like slope of the linear method's elimination XORs: 0.848 per bit over n ≤ 23 and 0.780 over all rungs. Both are still below F4's 1.079.

The rho slopes sit below 0.5 because the Frobenius fold divides rho's work by √(2n), and fixed per-solve costs dominate at the smallest rungs. They are reported as measured.




work against n




distance to rho

## §5a band

The IC's main work is unpriced native work: elimination word XORs plus about 3d field operations per trial. Its counted group additions come from anchor steps, factor-base projection and relation checks; rho's come from its walk. Per solve, from the rung records:

| a | n | IC group additions | IC elimination XORs | rho group additions | IC ÷ rho at 0 GAE | at 0.1 GAE | at 1 GAE |
|---|---|---|---|---|---|---|---|
| 0 | 13 | 376 | 14,040 | 74 | 5.1 | 24 | 195 |
| 0 | 23 | 3,193 | 5,039,027 | 674 | 4.7 | 753 | 7,481 |
| 0 | 41 | 4,484,130 | 72,513,368,117 | 96,270 | 47 | 75,370 | 753,276 |

Rho is cheaper at 0, 0.1 and 1 GAE per unpriced operation on every rung, so the order holds across the band. The rung records keep rho's group additions and walk steps but not its unpriced counters. Bounding those by three per walk step (one canonicalization, one partition hash, at most one table access) raises rho's figure at 1 GAE at most fourfold, far inside the margin (at least 195× at 1 GAE); the IC column also omits its uncounted field multiplications, which would only raise it. Neither changes the order.

## What this does and does not show

- It lowers the algebraic IC's fitted cost exponent from about 1.08 to about 0.65 per bit of n, and extends the largest completed algebraic solve from n = 23 to n = 41. This is a measured improvement over the algebraic baseline under the owner's criterion.
- It does not approach rho. Its floor exponent 2/3 is above rho's 1/2, so the distance grows with n ($\log_2$ slope of the ratio ≈ 0.47 per bit).
- **N83.** On the a = 0 curve (cofactor 4, 81-bit subgroup) the method would take about 2^{55} trials, against about 2^{37} rho steps. On the a = 1 curve (cofactor about $1.1 \cdot 10^9$) the relation graph fragments as at n = 37, so the method does not apply as built.
- **Next levers.** Lowering the exponent below 2/3 needs new algebra. Symmetric S4 in (e1, e2, e3) has four nonlinear monomials ($s3^3$, $s2^2s3$, $s1^2s3$, $s1^2s3^2$), so three summands do not linearize the same way. The large-cofactor case needs a factor base or relation scheme that does not split by cofactor class.

## Validation

- `replay-certificate.json`: `verified`. Each rung was re-executed in a fresh process (all targets, n = 41 sampled at target 0), and every integer field matched the record. All 104 IC and rho scalars were checked with `KoblitzCurve::mul` (BigUint), separate from the producer's field code.
- `validator-report.json`: `pass`. An independent validator agent recomputed all 18 headline numbers with jq/awk from the raw rung files, without reading the fit or replay code; differences were zero. It also confirmed every rung's h·r against the curve order and judged the method algebraic under the owner's criterion. Its caveats: the $2^{12}$ cap on solution enumeration can miss relations but never yield wrong ones; the F4 comparison mixes units, as above; and the method depends on the cofactor.

- The producer was frozen at `995b187c1`. Its source in this PR differs only by rustfmt layout and three clippy fixes: an unused helper removed, an uninitialized `let`, and `is_multiple_of`. None of these changes behaviour.

## Files

`PROTOCOL.md` · `raw/` (rung records, logs, orchestration, binary digest, operator notes) · `fit.json` (`examples/ic_m2_ladder_fit.rs`) · `replay-certificate.json` (`examples/ic_m2_ladder_replay.rs`) · `validator-report.json` (independent validator) · `VALIDATION.json` · `plot.py`, `work_vs_n.svg/png`, `distance_to_rho.svg/png` · `REPORT.pdf` · producer `examples/ic_m2_linear.rs`.